

LUODIAN, China

WMO: 579160

Lat: 25.4286N

Lon: 106.7638E

Elev: 441

StdP: 96.14

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.3	(c) 4.6	(d) -2.8	(e) 3.2	(f) 7.8	(g) -1.0	(h) 3.7	(i) 8.4	(j) 3.7	(k) 14.8	(l) 3.2	(m) 13.6	(n) 0.7	(o) 140	(p) 0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	35.1	25.3	34.2	25.3	33.2	25.0	26.9	32.7	26.4	31.9	26.0	31.2	1.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	21.9	29.5	25.1	21.3	29.0	24.7	20.8	28.5	88.1	32.9	85.9	32.1	84.0	31.3	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.4	2.9	2.4		1.2	37.1	1.3	0.9	0.2	37.8	-0.5	38.3	-1.2	38.8	-2.2	39.4	
			WB	0.4	27.9	1.4	0.6	-0.6	28.3	-1.4	28.6	-2.2	28.9	-3.2	29.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	10.9	13.5	17.4	22.2	24.8	26.5	27.8	27.7	25.5	21.7	17.4	12.5
	DBStd	6.70	3.89	4.80	4.42	4.01	2.91	2.11	1.59	1.76	2.55	2.98	3.29	2.97
	HDD10.0	60	34	15	1	0	0	0	0	0	0	0	1	9
	HDD18.3	707	231	149	73	11	0	0	0	0	0	6	56	181
	CDD10.0	3966	63	114	231	366	458	495	552	548	466	364	223	87
	CDD18.3	1573	2	14	44	127	201	245	293	289	216	112	28	2
	CDH23.3	13597	13	127	321	1020	1688	2182	2938	2782	1757	623	135	12
(13)	CDH26.7	5326	1	39	109	408	654	811	1209	1199	692	179	21	2
(14) Wind	WSAvg	0.8	0.8	0.9	1.0	1.0	0.9	0.7	0.7	0.7	0.7	0.6	0.6	0.7
(15) Precipitation	PrecAvg	1150	15	19	31	93	208	211	189	161	76	87	45	17
	PrecMax	1625	43	51	127	252	563	418	460	291	188	257	167	65
	PrecMin	775	0	2	1	17	62	62	22	50	25	16	1	0
	PrecStd	228	9	11	27	55	106	89	109	67	44	65	38	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	31.1	33.0	36.0	35.7	35.2	35.9	35.9	35.5	32.1	29.2	25.2
		MCWB	17.4	18.8	20.7	22.0	24.8	26.4	26.3	25.7	25.0	22.5	21.0	18.0
	2%	DB	21.6	27.2	29.7	33.4	33.7	33.7	34.5	34.8	33.6	30.3	26.6	21.2
		MCWB	15.6	18.3	20.5	22.7	24.5	25.9	26.2	25.7	24.6	22.4	19.8	15.3
	5%	DB	18.8	23.9	27.0	31.2	32.1	32.4	33.4	33.7	32.3	28.7	24.8	19.1
		MCWB	13.7	17.2	19.5	22.5	23.8	25.4	25.9	25.6	24.1	21.7	18.8	13.9
	10%	DB	16.3	20.8	24.2	28.9	30.4	31.0	32.2	32.4	30.7	27.0	22.9	17.1
		MCWB	12.6	15.7	18.1	21.5	23.2	25.0	25.7	25.4	23.4	21.1	17.9	12.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	19.9	22.1	24.6	26.6	27.5	27.6	27.2	26.4	24.6	22.0	18.2
		MCDB	24.2	28.0	29.6	31.9	33.1	33.5	33.4	33.2	32.6	29.8	26.6	23.7
	2%	WB	16.0	18.5	20.8	23.7	25.6	26.7	26.9	26.6	25.6	23.7	20.9	16.4
		MCDB	21.0	25.6	27.9	30.6	31.7	32.1	32.6	32.4	31.3	28.3	24.9	20.1
	5%	WB	14.6	17.3	19.9	22.9	24.8	26.1	26.4	26.1	25.1	22.8	19.9	15.0
		MCDB	17.8	23.4	26.0	29.1	30.3	30.9	31.7	31.9	30.2	27.0	23.9	18.0
	10%	WB	13.1	16.1	18.8	22.1	24.1	25.6	26.0	25.7	24.5	22.0	18.7	13.6
		MCDB	15.7	20.6	23.7	27.7	28.7	29.7	30.8	30.9	28.9	25.4	21.8	16.4
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.6	8.2	9.0	8.6	7.5	8.0	8.8	8.9	7.6	7.9	7.1
		MCDBR	12.1	14.3	13.0	12.6	11.7	9.9	10.3	11.1	11.5	10.8	11.3	12.0
		MCWBR	5.9	5.7	4.7	4.0	3.9	3.2	3.2	3.1	3.3	3.6	4.8	6.1
	5% WB	MCDBR	10.8	13.1	12.2	11.2	9.4	8.4	9.0	9.8	9.5	8.8	9.5	10.1
		MCWBR	5.5	5.4	4.8	4.1	3.7	3.0	3.1	3.1	3.0	3.3	4.4	5.5
(40) Clear-Sky Solar Irradiance	taub		0.619	0.621	0.746	0.708	0.584	0.566	0.525	0.579	0.596	0.680	0.622	0.648
	taud		1.612	1.644	1.453	1.570	1.893	1.962	2.082	1.912	1.837	1.600	1.656	1.572
	Ebn at Noon		621	668	613	656	743	751	782	740	713	614	614	573
	Edh at Noon		232	241	306	278	201	186	165	195	205	248	220	234
(44) All-Sky Solar Radiation	RadAvg		1.79	2.59	3.17	4.03	4.20	3.83	4.67	4.54	3.91	2.82	2.62	1.96
	RadStd		0.46	0.54	0.42	0.51	0.48	0.50	0.34	0.33	0.39	0.40	0.39	0.33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A	+49

Nomenclature: See separate page